

Assignment Day – 5

1. A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.



Step	Troubleshooting Step	What I am checking
1	Check the power adapter and cable	Check for loose connections, damaged cables, or a faulty charger.
2	Check the power outlet	Plug in another device to confirm that the wall outlet is providing power.
3	Check the charging port	Look for dust, damage, or a loose charging port that may prevent power from reaching the laptop.
4	Perform a power reset	Disconnect the charger, remove the battery if removable, hold the power button for 15–30 seconds, then reconnect power. This clears residual electrical charge.
5	Check the battery and internal hardware	Check whether the battery is faulty and inspect internal power connections or motherboard damage if the laptop still shows no signs of power.

2. You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.



Monitor Blank Screen – Troubleshooting Flow

1. **Check the monitor power** → Make sure the monitor is switched on and the power LED is lit.
2. **Check the display cable** → Ensure the HDMI/DisplayPort cable is firmly connected to both the monitor and PC.
3. **Check the correct input source** → Select the correct HDMI/DisplayPort input on the monitor.
4. **Check the GPU connection** → If the PC has a dedicated GPU, make sure the display cable is connected to the **GPU port**, not the motherboard port.
5. **Try another cable or monitor** → This helps determine whether the cable or monitor is faulty.
6. **Check Windows display settings** → Press **Windows + P** and select **Duplicate** or **Extend**. Also try **Windows + Ctrl + Shift + B** to reset the graphics driver.
7. **Check the GPU and RAM** → Power off the PC, reseal the GPU and RAM, and check for loose connections.
8. **Restart and confirm** → Boot the PC again and check whether the display returns.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.



- **Check the mouse power** – Make sure the mouse is switched on and the battery is charged or replaced. This checks whether the mouse itself has power.
- **Check the USB receiver** – Remove and reconnect the wireless mouse receiver. Make sure it is securely connected.
- **Try another USB port** – Plug the receiver into a different USB port. If it works, the original USB port may have a problem.
- **Test the mouse on another computer** – If the mouse still does not work, connect it to another PC. If it fails there too, the mouse or receiver is likely faulty.
- **Check system settings and drivers** – In **Device Manager**, check under **Mice and other pointing devices** or **Human Interface Devices** for errors. Update or reinstall the mouse driver if necessary.
- **Check for interference** – Keep the mouse receiver away from other wireless devices and USB 3.0 devices that may cause interference.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hint: Think about fans, dust, and thermal paste.



Hardware Cause	How I Would Check
1. Faulty or slow CPU/GPU fan	Check whether the fans are spinning properly during operation. Listen for unusual noise and check temperatures using a hardware-monitoring tool.
2. Dust buildup	Power off and unplug the PC, open the case, and inspect the fans, heatsinks, and air vents. Clean accumulated dust carefully using compressed air.
3. Old or dried thermal paste	Inspect the CPU/GPU heatsink. If the thermal paste is dried or cracked, clean the old paste and apply fresh thermal paste before reinstalling the heatsink properly.